

Can we put a price on climate impacts? *Measuring and valuing damages*

ECON-4210

Economics of Climate Change

Class 11

Prof. Casey Wichman

Today's class

1. Questions?

2. Attendance

3. Housekeeping

1. Fall Break next week (no class on Monday)
2. Exam 1

4. For today

1. Estimating climate damages (Aggregate GDP + effects on agriculture)

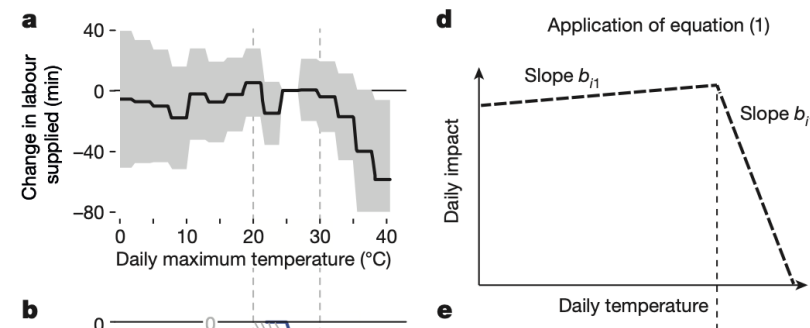
Global non-linear effect of temperature on economic production

Marshall Burke^{1,2*}, Solomon M. Hsiang^{3,4*} & Edward Miguel^{4,5}

Growing evidence demonstrates that climatic conditions can have a profound impact on the functioning of modern human societies^{1,2}, but effects on economic activity appear inconsistent. Fundamental productive elements of modern economies, such as workers and crops, exhibit highly non-linear responses to local temperature even in wealthy countries^{3,4}. In contrast, aggregate macroeconomic productivity of entire wealthy countries is reported not to respond to temperature⁵, while poor countries respond only linearly^{5,6}. Resolving this conflict between micro and macro observations is critical to understanding the role of wealth in coupled human–natural systems^{7,8} and to anticipating the global impact of climate change^{9,10}. Here we unify these seemingly contradictory results by accounting for non-linearity at the macro scale. We show that overall economic productivity is non-linear in temperature for all countries, with productivity peaking at an annual average temperature of 13 °C and declining strongly at higher temperatures. The relationship is globally generalizable, unchanged since 1960, and apparent for agricultural and non-agricultural activity in both rich and poor countries. These results provide the first evidence that economic activity in all regions is coupled to the global climate and establish a new empirical foundation for modelling economic loss in response to climate change^{11,12}, with important implications. If future adaptation mimics past

wealthy populations actually are unaffected by temperature, this could indicate that wealth and human-made capital are substitutes for natural capital (for example, the composition of the atmosphere) in economic activity^{5,7}. Resolving this apparent discrepancy thus has central implications for understanding the nature of sustainable development⁷.

Numerous basic productive components of an economy display a highly non-linear relationship with daily or hourly temperature¹. For example, labour supply⁴, labour productivity⁶, and crop yields³ all decline abruptly beyond temperature thresholds located between 20 °C and 30 °C (Fig. 1a–c). However, it is unclear how these abrupt



Global economic damages

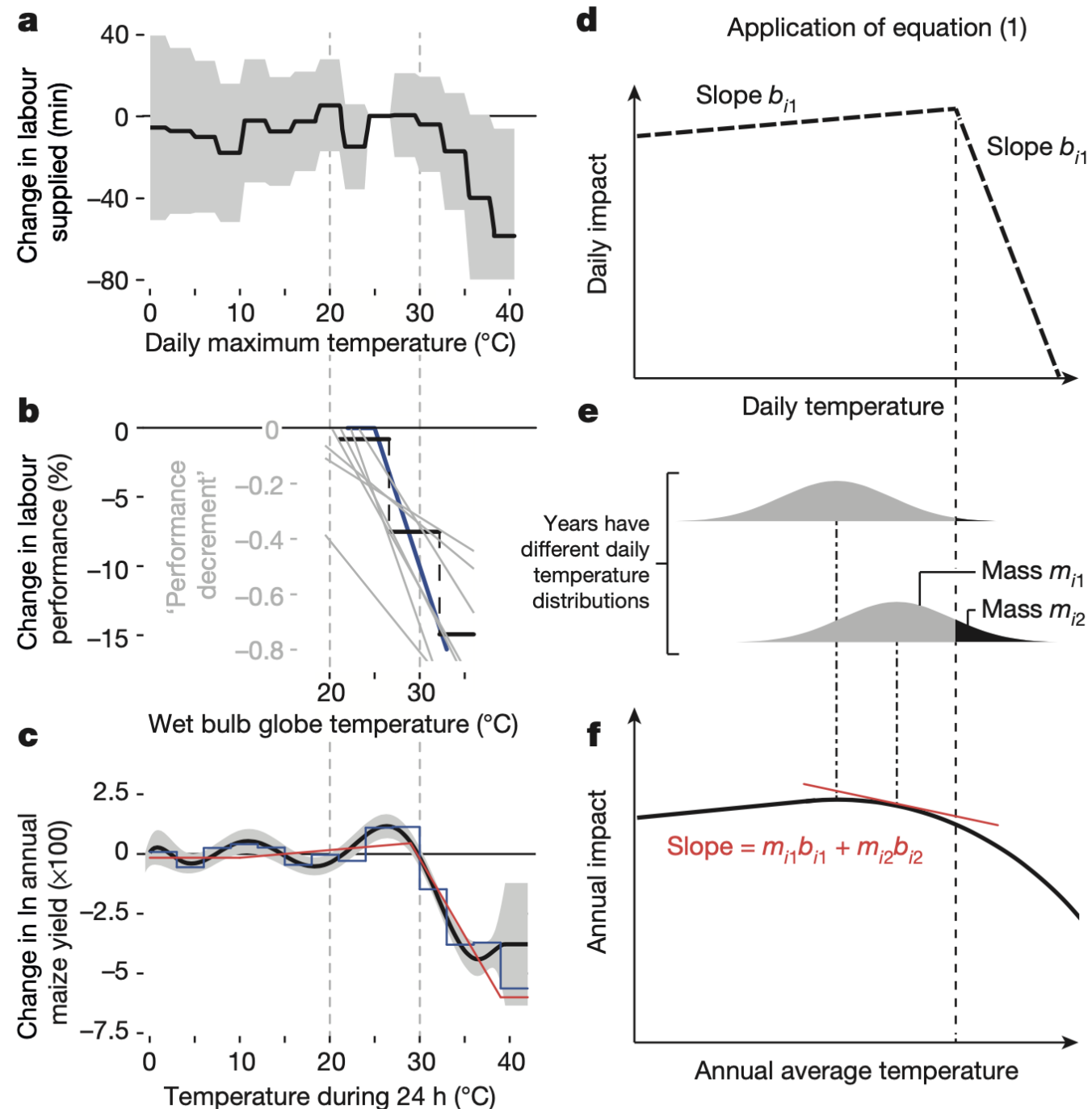
Accounting for “nonlinearities” a la Burke, Hsiang, & Miguel (2015)

- “Micro” estimates in individual sectors seem to imply that extreme temperatures are bad for crop yields, labor supply and productivity, mortality, etc.
- But, “macro” studies tend to show a little/no relationship between economic output and temperature for rich countries and a linear relationship for poor countries.
- This is, effectively focusing on the shape of f in a simple equation we’ve seen before:

$$\Delta GDP = f(T)$$

From micro to macro weather responses

- Highly nonlinear micro responses to temperature (panel a-c) can be aggregated up to a “smooth” concave macro response (panel f)
- There are sector-specific “thresholds” beyond which damages become worse quite quickly (E.g., $>30^{\circ}\text{C}$ for maize yield).
 - We’ll come back to where this comes from later today!



Empirical strategy

Burke, Hsiang, Miguel (2015)

- Data:
 - ▶ 166 countries from 1960-2010.
- “Ideal experiment”
 - ▶ Take two identical countries, warm the temperature of one of them over 100+ years, compare economic output.
 - ▶ The authors simulate this design by comparing economic output for a given country when they face hotter/colder weather in any given year.
- Econometric design:
 - ▶ Outcome variable is year-to-year economic growth rates (changes in $\ln(\text{GDP per capita})$)
 - ▶ Controlling for: country and year “fixed effects” (e.g., shocks over time that affect all countries), quadratic time trends for each country.
 - ▶ Focused on estimating the shape of the weather-response function for average annual temperature.

B.2 Empirical approach

From: Burke, Hsiang, & Miguel (2015) supplementary materials...

Using a 51-year longitudinal sample of countries around the world, we take first differences of the natural log of annual real (inflation-adjusted) gross domestic product per capita. The first difference of the natural log of annual real (inflation-adjusted) gross domestic product per capita (ΔY) can be interpreted as per-period growth rates in income. We decompose these changes in income with a simple and general model:

**Time controls
& fixed effects**

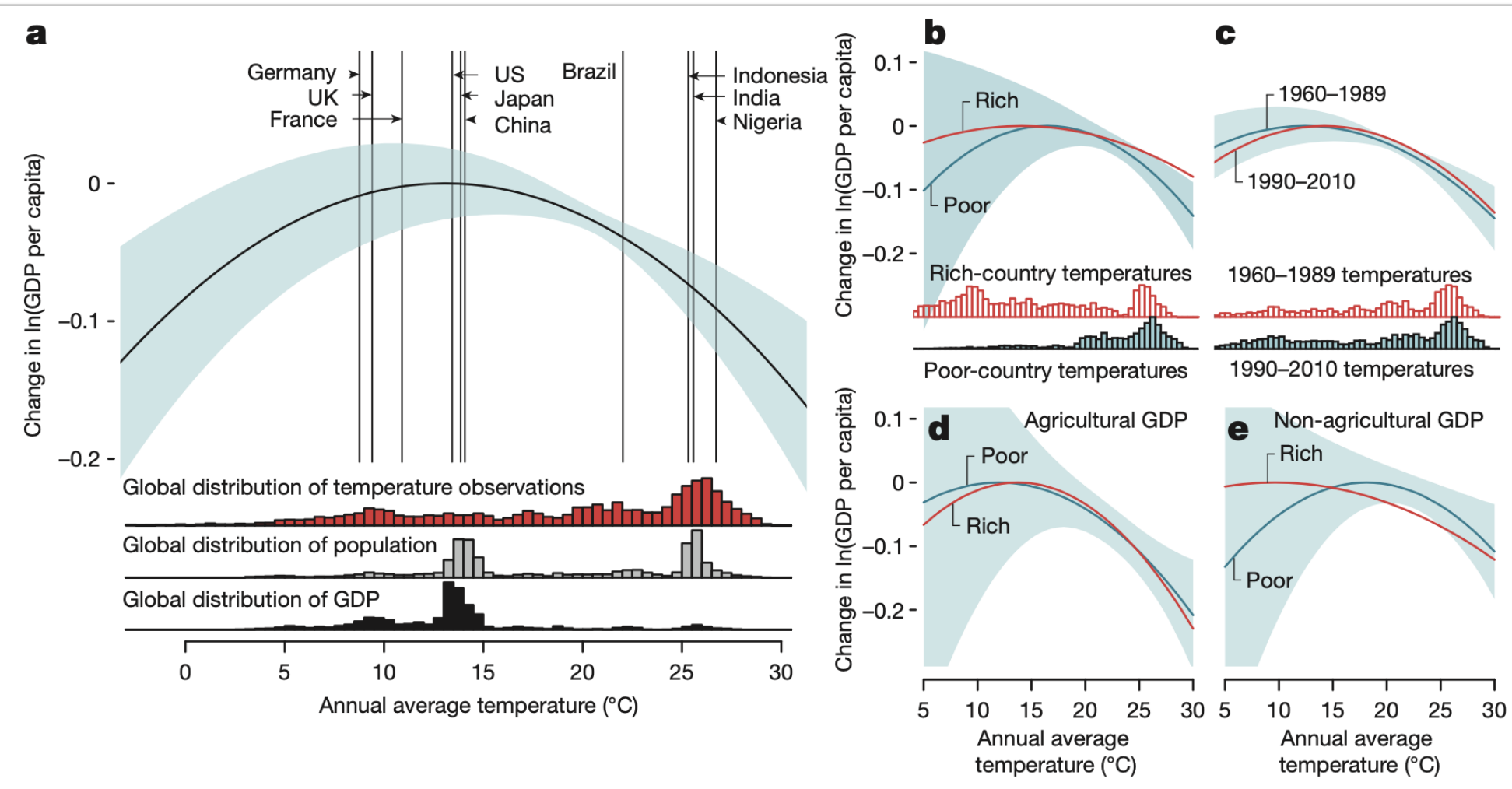
$$\Delta Y_{it} = h(T_{it}) + \lambda_1 P_{it} + \lambda_2 P_{it}^2 + \mu_i + \nu_t + \theta_{it} + \theta_{i2}t^2 + \varepsilon_{it} \quad (15)$$

**Weather-response
function**

where countries are indexed by i and years by t . Note that these definitions, used through the remainder of the SOM, differ from the definitions for i and t used in Sections A.1-A.3. All time-invariant factors that influence countries' average growth rates, such as history, culture or topography, are accounted

Our focus is the potentially non-linear relationship between annual temperatures T_{it} and income growth described by $h(T_{it})$. We begin by estimating $h(T_{it})$ as a simple quadratic (i.e. $h(T_{it}) = \beta_1 T_{it} + \beta_2 T_{it}^2$), exploring more flexible functional forms such as higher order polynomials and restricted cubic splines for robustness.

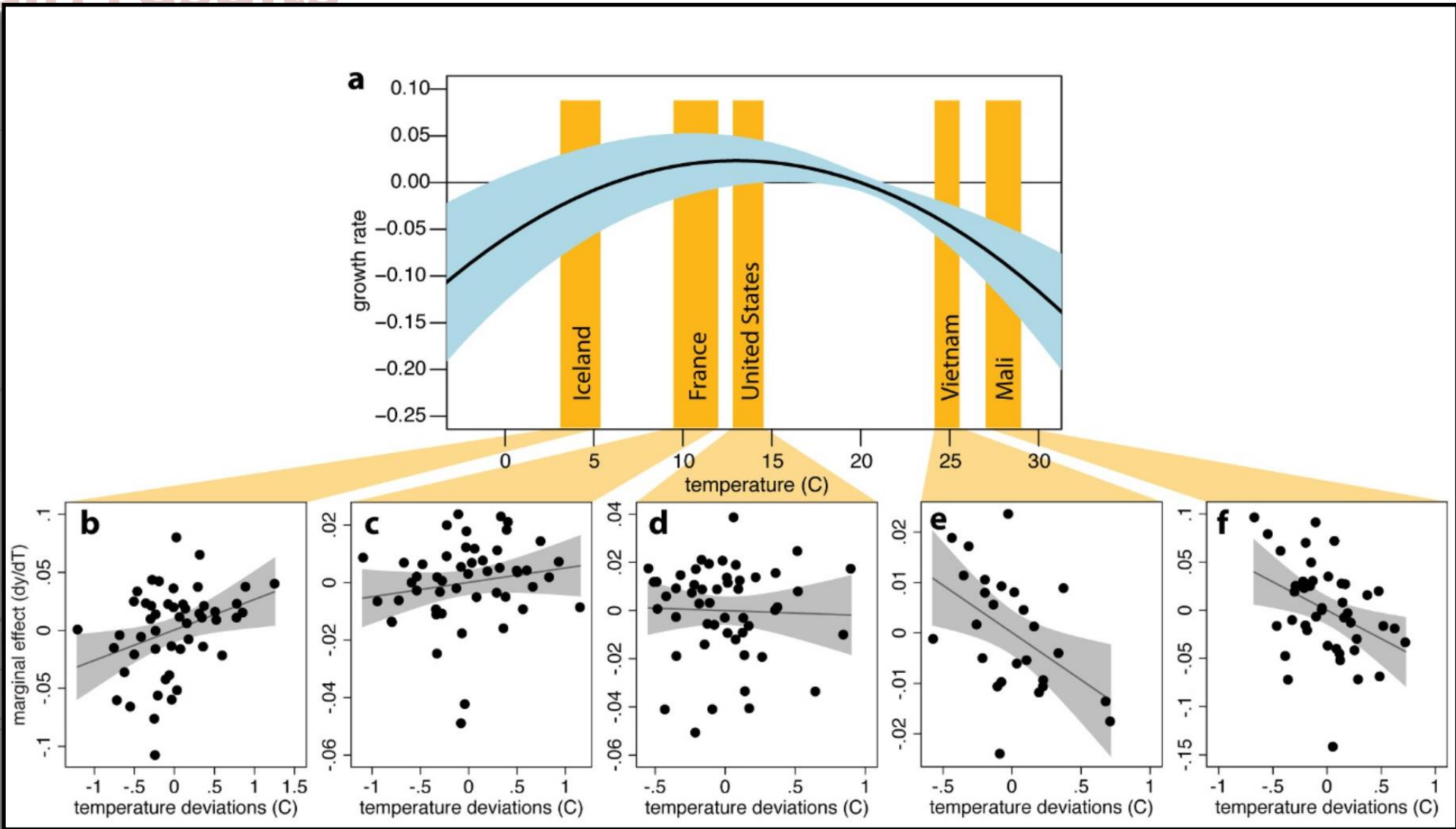
Main results



- Inverted U-shaped relationship between change in $\ln(\text{GDPpc})$
- Similar effects for rich/poor countries (panel b)
- Most rich countries are clustered around 13-15 $^{\circ}\text{C}$.
- Relationship hasn't changed between 1960-1989 and 1990-2010 (panel c)
- Inverted-U shape exists for ag and non-ag GDP (panel d & e)
- **Takeaway:** only weak evidence that rich countries are less vulnerable to climate change. Many poorer countries are located further "right" in the temperature distribution and suffer more damage as a result.

Main results

- Inverted U-shaped relationship between change in $\ln(\text{GDPpc})$



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Main results

The modeling may seem complicated (and, in a variety of ways, it is), but the main results just stem from these coefficients on temperature and temperature-squared!

Extended Data Table 1 | Regression estimates for global sample, main estimate and robustness

	(1) Base	(2) >20yrs	(3) No oil	(4) No US/China	(5) ContYr FE	(6) ContYr + noTrend	(7) No YrFE	(8) LinearTime	(9) LDV 1lag	(10) LDV 3lags	(11) PWT
Temp.	0.0127*** (0.0038)	0.0135*** (0.0038)	0.0128*** (0.0036)	0.0128*** (0.0038)	0.0142*** (0.0037)	0.0133*** (0.0034)	0.0103*** (0.0039)	0.0128*** (0.0043)	0.0087** (0.0039)	0.0062* (0.0038)	0.0072* (0.0039)
Temp. sq.	-0.0005*** (0.0001)	-0.0005*** (0.0001)	-0.0005*** (0.0001)	-0.0005*** (0.0001)	-0.0005*** (0.0001)	-0.0004*** (0.0001)	-0.0004*** (0.0001)	-0.0005*** (0.0001)	-0.0004*** (0.0001)	-0.0003*** (0.0001)	-0.0004*** (0.0001)
Precip.	0.0145 (0.0100)	0.0148 (0.0100)	0.0130 (0.0101)	0.0148 (0.0101)	0.0124 (0.0106)	0.0084 (0.0098)	0.0159 (0.0107)	0.0137 (0.0100)	0.0165* (0.0095)	0.0201** (0.0098)	0.0195* (0.0109)
Precip. sq.	-0.0047* (0.0026)	-0.0049* (0.0026)	-0.0040 (0.0025)	-0.0048* (0.0026)	-0.0041 (0.0027)	-0.0021 (0.0023)	-0.0045* (0.0026)	-0.0035 (0.0023)	-0.0047* (0.0024)	-0.0052** (0.0024)	-0.0038 (0.0028)
Constant	1.4575** (0.6444)	0.0740 (0.0633)	1.4522** (0.6228)	1.4707** (0.6507)	-0.0362 (0.0411)	-0.0819** (0.0370)	-0.8024** (0.3366)	-0.7693*** (0.0517)	-11.3227*** (0.7957)	-28.3451*** (2.0763)	0.0643 (0.0467)
Observations	6584	6477	6090	6484	6584	6584	6584	6584	6418	6086	6627
R squared	0.286	0.278	0.275	0.284	0.367	0.267	0.240	0.219	0.286	0.289	0.220
Optimum	13.06	13.39	12.64	13.09	14.92	17.40	12.74	13.40	11.92	9.98	9.88

Unless otherwise indicated, all models include country fixed effects, year fixed effects, and quadratic country time trends, with errors clustered at the country level. Temperature is measured in °C and precipitation in metres. Columns: (1) main specification, (2) as in column 1 but excluding countries with fewer than 20 years of growth data, (3) as in column 1 but dropping large oil exporting countries, (4) as in column 1 but dropping United States and China, (5) as in column 1 but adding continent-by-year fixed effects, (6) as in column 1 but adding continent-by-year fixed effects and dropping country time trends, (7) as in column 1 but dropping year fixed effects, (8) as in column 1 but only linear time trend, (9–10) as in column 1 but adding 1 or 3 lags of per capita growth (that is, lagging the dependent variable), (11) as in column 1 but using growth data from Penn World Tables. Asterisks indicate statistical significance at the 1% (***), 5% (**), and 10% (*) levels.

Annual average temperature (°C)

5 10 15 20 25 30 5 10 15 20 25 30

Annual average
temperature (°C)

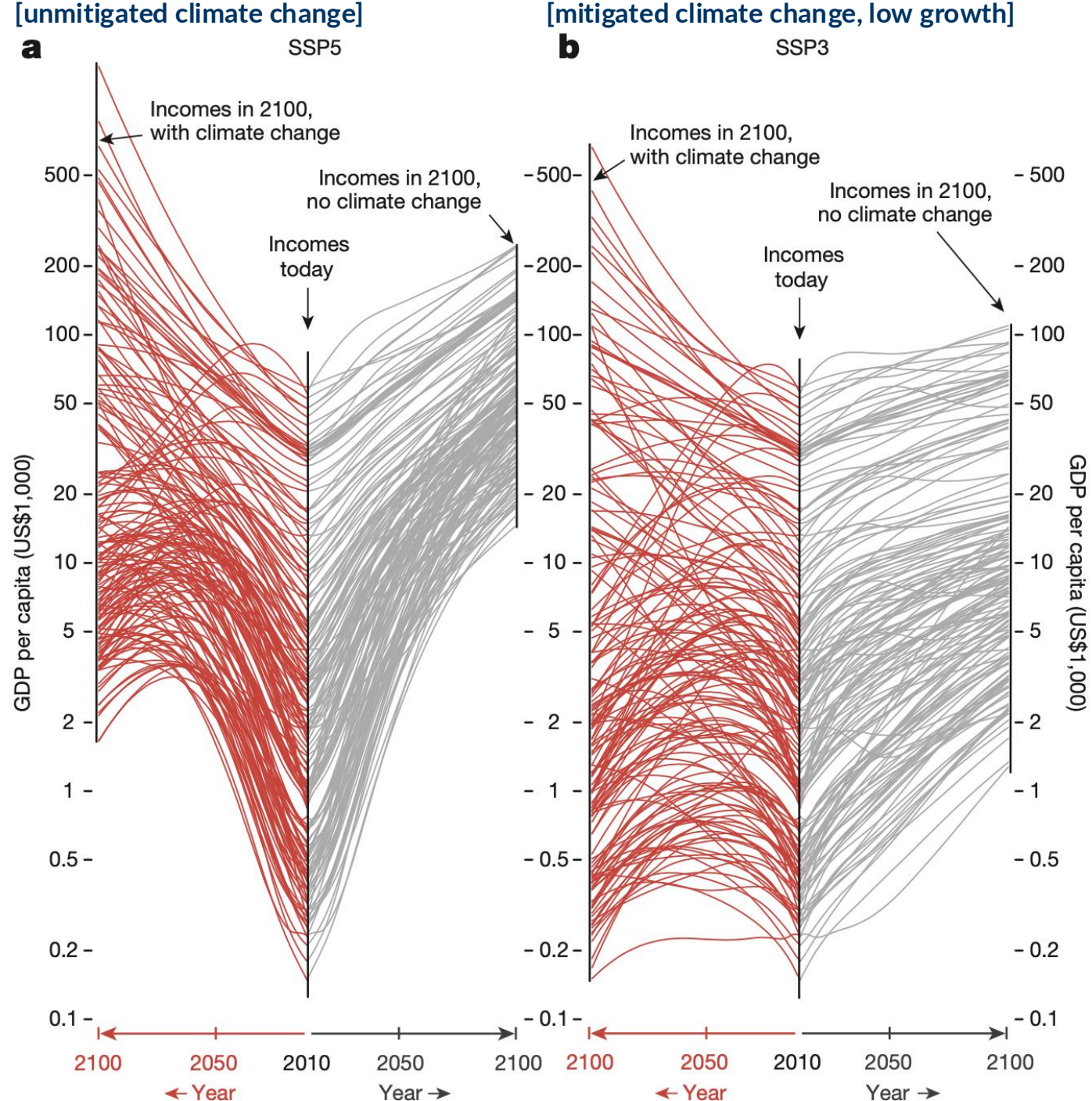
Annual average
temperature (°C)

How will climate affect countries' GDP?

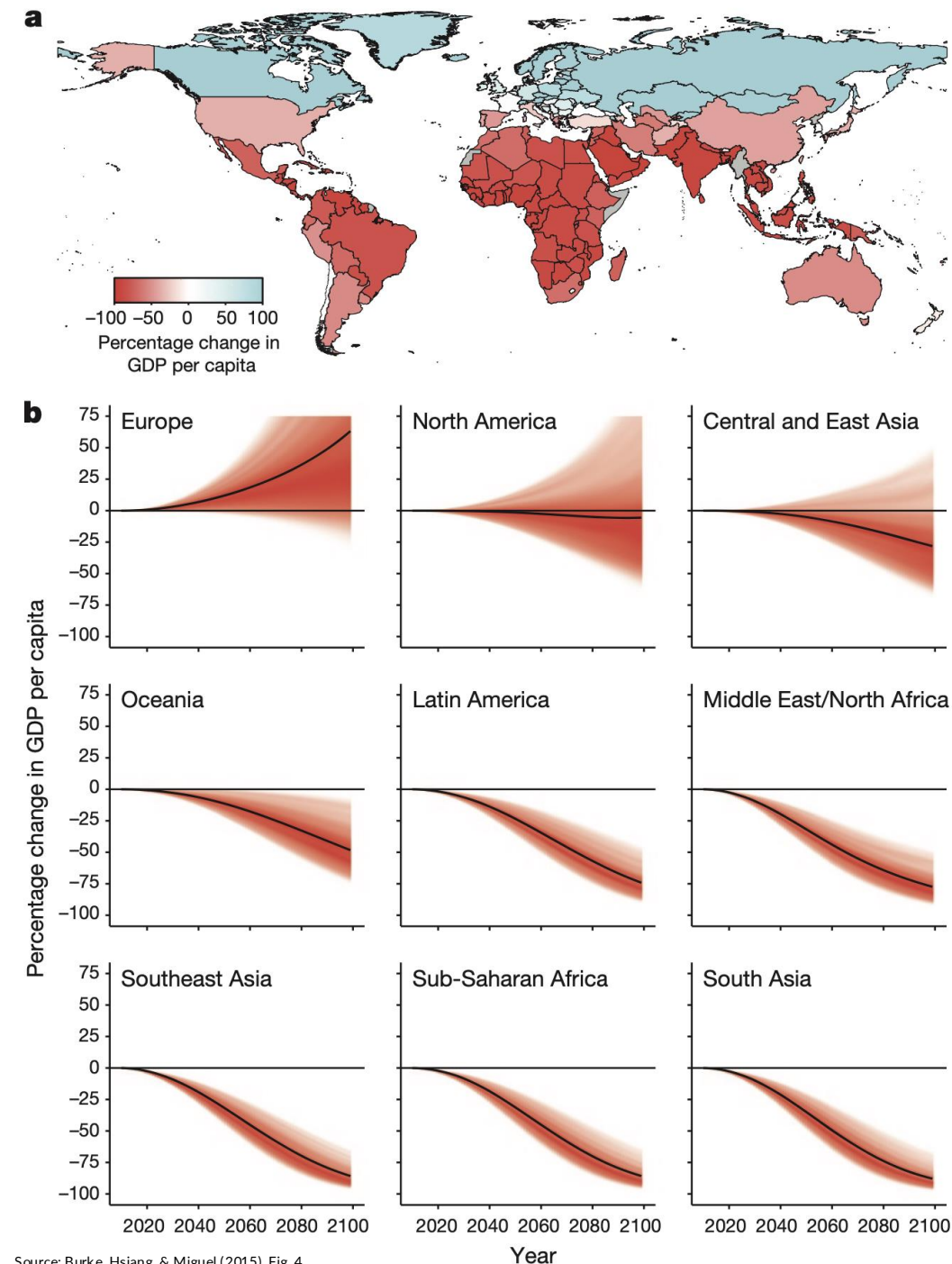
- 77% of countries will be poorer in 2100 with climate change relative to a “no climate change” scenario (under SSP5)
 - 5% poorer in 2100 than they are today (SSP5)
 - 43% poorer in 2100 than they are today (SSP3)
- Even with unmitigated climate change (SSP5) many countries become wealthier (in per capita terms)

Socioeconomic scenarios:

- SSP3 — high challenges to mitigation and adaptation.
- SSP5 — fossil-fuel development (strong economic growth, but little GHG mitigation)



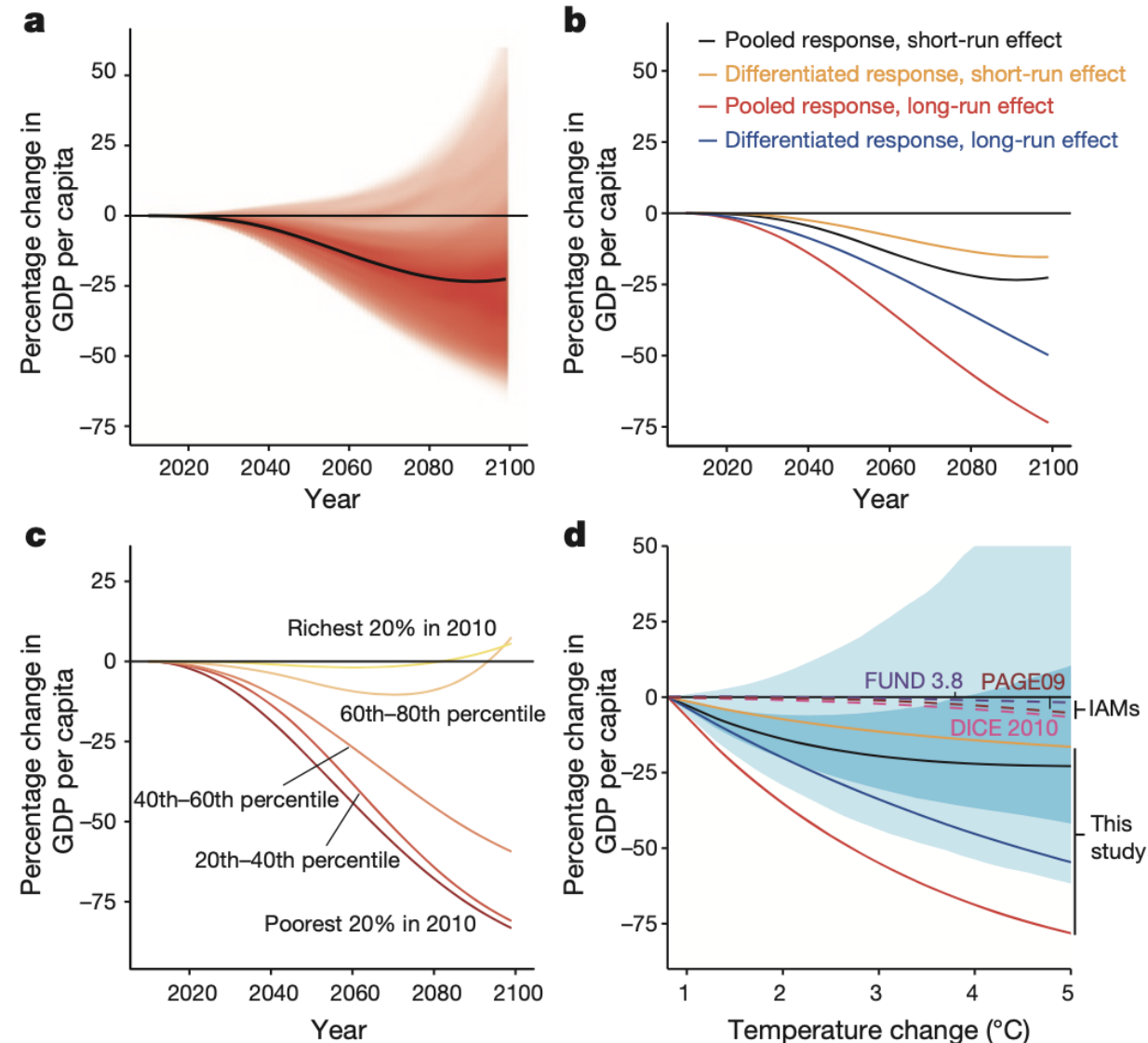
Regional effects of temperature change



- Large changes in GDP by 2100 in: Oceania, Central and South America, Middle East, Africa, South and Southeast Asia...
- Smaller (with wide confidence intervals) in North America.
- Increased temperatures lead to GDP growth in Europe.
- One implication: using climate impacts from the US (where we have good data) is likely not informative for countries that will be more heavily impacted by climate change.

Income heterogeneity and comparison to SCC-IAMS

- Primary results suggest a 25% loss in GDP per capita by 2100 (panel a).
- Poorest countries are expected to see worst outcomes.
- Empirical global damage function (panel d) is somewhat linear (and slightly concave).
 - Suggests much greater damage at 2-3°C than SCC-IAMs
 - Only includes damages from average annual temperature changes and not, e.g., sea-level rise, cyclones, etc.
 - Remember: DICE, FUND, & PAGE are the three IAMs that were previously used by the US Gov't to estimate the SCC.



Takeaways on Burke et al. (2015)

- If future adaptation mimics past adaptation, unmitigated warming will reduce global incomes by roughly 23%
 - This is somewhat different from the 2-3% of GDP estimated in the (now outdated) SCC-IAMs...
- Much of the difference comes from Burke et al. modeling damages to GDP growth rates, rather than GDP levels.
 - Effects on growth rates compound over time, making this effect much larger than previous estimates.
 - Active source of debate in climate economics!
- Rich and poor countries both vulnerable to climate change. Poorer countries tend to be hotter countries, so will likely experience economic damages to GDP growth sooner.
- Empirically estimated damage functions suggest much bigger damages from climate change than do IAMs used to estimate the SCC.
 - Striking because only average temperatures are considered, and so many other climate impacts are omitted.

Nonlinear temperature effects indicate severe damages to U.S. crop yields under climate change

Wolfram Schlenker^{a,1} and Michael J. Roberts^b

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Communicated by V. Kerry Smith, Arizona State University, Tempe, AZ, July 1, 2009 (received for review October 13, 2008)

The United States produces 41% of the world's corn and 38% of the world's soybeans. These crops comprise two of the four largest sources of caloric energy produced and are thus critical for world food supply. We pair a panel of county-level yields for these two crops, plus cotton (a warmer-weather crop), with a new fine-scale weather dataset that incorporates the whole distribution of temperatures within each day and across all days in the growing season. We find that yields increase with temperature up to 29° C for corn, 30° C for soybeans, and 32° C for cotton but that temperatures above these thresholds are very harmful. The slope of the decline above the optimum is significantly steeper than the incline below it. The same nonlinear and asymmetric relationship is found when we isolate either time-series or cross-sectional variations in temperatures and yields. This suggests limited historical adaptation of seed varieties or management practices to warmer temperatures because the cross-section includes farmers' adaptations to warmer climates and the time-series does not. Holding current growing regions fixed, area-weighted average yields are predicted to decrease by 30–46% before the end of the century under the slowest (B1) warming scenario and decrease by 63–82% under the most rapid warming scenario (A1FI) under the Hadley III model.

agriculture | panel analysis | time series | cross section | farmer adaptation

days of the growing season, all estimated for the specific locations within each county where crops are grown. The new fine-scale weather data facilitate estimation of a flexible model that can detect nonlinearities and breakpoints in the effect of temperature on yield. If the true underlying relationship is nonlinear (e.g., increasing and then decreasing in temperature), averaging over time or space dilutes the true temperature response. For example, similar average temperatures may arise from two very different days, one with little temperature variation and one with wide temperature variation. Holding the average temperature constant, days with more variation will include more exposure to extreme outcomes, which can critically influence yields (9). The empirical challenge is to map an entire season of widely varying temperature outcomes to each year's yield. Accurate estimation of nonlinear effects is particularly important when considering large, nonmarginal changes in temperatures, now expected with climate change.

Results

Regression analyses, which control in various ways for precipitation, technological change, soils, and location-specific unobserved factors, all show a similar nonlinear relationship between temperature and yields. Additional results and sensitivity checks are given in the [SI Appendix](#).

- Switching gears to focus on agricultural damages from temperature extremes.
- Agricultural is (perhaps obviously) very dependent on climate.
- This paper focuses on corn and soybeans, of which the US contributes 41% and 38% of the world's supply.
- Extreme temperatures can lead to severe damages to agricultural productivity, which would have ripple effects throughout the world economy.

Schlenker & Roberts (2009) approach

- Data:

- ▶ Crop yields from 1950-2005 for most US counties.
- ▶ Fine-scale weather data (number of days each crop experiences extreme temperature).

- Empirical framework:

- ▶ Challenge is that weather data is daily max/min but crop yields are annual.
- ▶ Use 3 flexible ways to model the distribution of temperature within a county.

This is just a complicated $y = f(T)$ -style equation where f is structured to capture the distribution of how many days each county experiences a certain temperature each year

temperature effects are additively substitutable over time. Specifically, plant growth $g(h)$ depends nonlinearly on heat h . Log yield, y_{it} , in county i and year t is

$$y_{it} = \int_{\underline{h}}^{\bar{h}} g(h) \phi_{it}(h) dh + \mathbf{z}_{it} \delta + c_i + \epsilon_{it} \quad [1]$$

where $\phi_{it}(h)$ is the time distribution of heat over the growing season in county i and year t . We fix the growing season to months March through August for corn and soybeans and the months April through October for cotton. Observed temperatures during this time period range between the lower bound $\underline{h}$ and the upper bound $\bar{h}$. Other control factors are denoted $\mathbf{z}_{it}$ and include a quadratic in total precipitation as well as quadratic time trends by state to capture technological change. Finally, c_i is a time-invariant county-fixed effect to control for time-invariant heterogeneity, such as soil quality. We allow the

Schlenker & Roberts (2009): Primary results

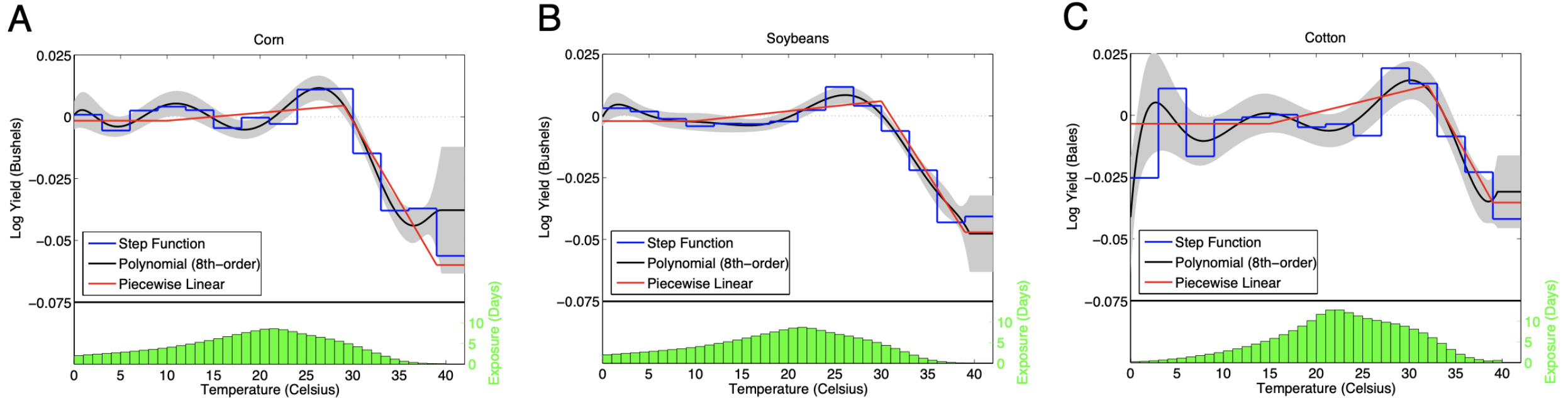


Fig. 1. Nonlinear relation between temperature and yields. Graphs at the top of each frame display changes in log yield if the crop is exposed for one day to a particular 1° C temperature interval where we sum the fraction of a day during which temperatures fall within each interval. The 95% confidence band, after adjusting for spatial correlation, is added as gray area for the polynomial regression. Curves are centered so that the exposure-weighted impact is zero. Histograms at the bottom of each frame display the average temperature exposure among all counties in the data.

Source: Schlenker & Roberts (2009), Fig. 1

- Interpretation (in A): exchanging one day at 29°C for one day at 40°C within the growing season would result in a $\approx 7\%$ reduction in corn yield.
- Different functional forms for the dose-response function produce similar results.

Climate impacts

- By end of the century, corn and soybean yields will:
 - go down by 30-46% in an optimistic climate scenario
 - go down by 63-82% in a pessimistic scenario.
- Counties in the north and south have similar weather-response functions.
- Weather-response function was similar in early years (1950-1977) and later years (1978-2005).
- Both of these suggest that there has been limited historical adaptation

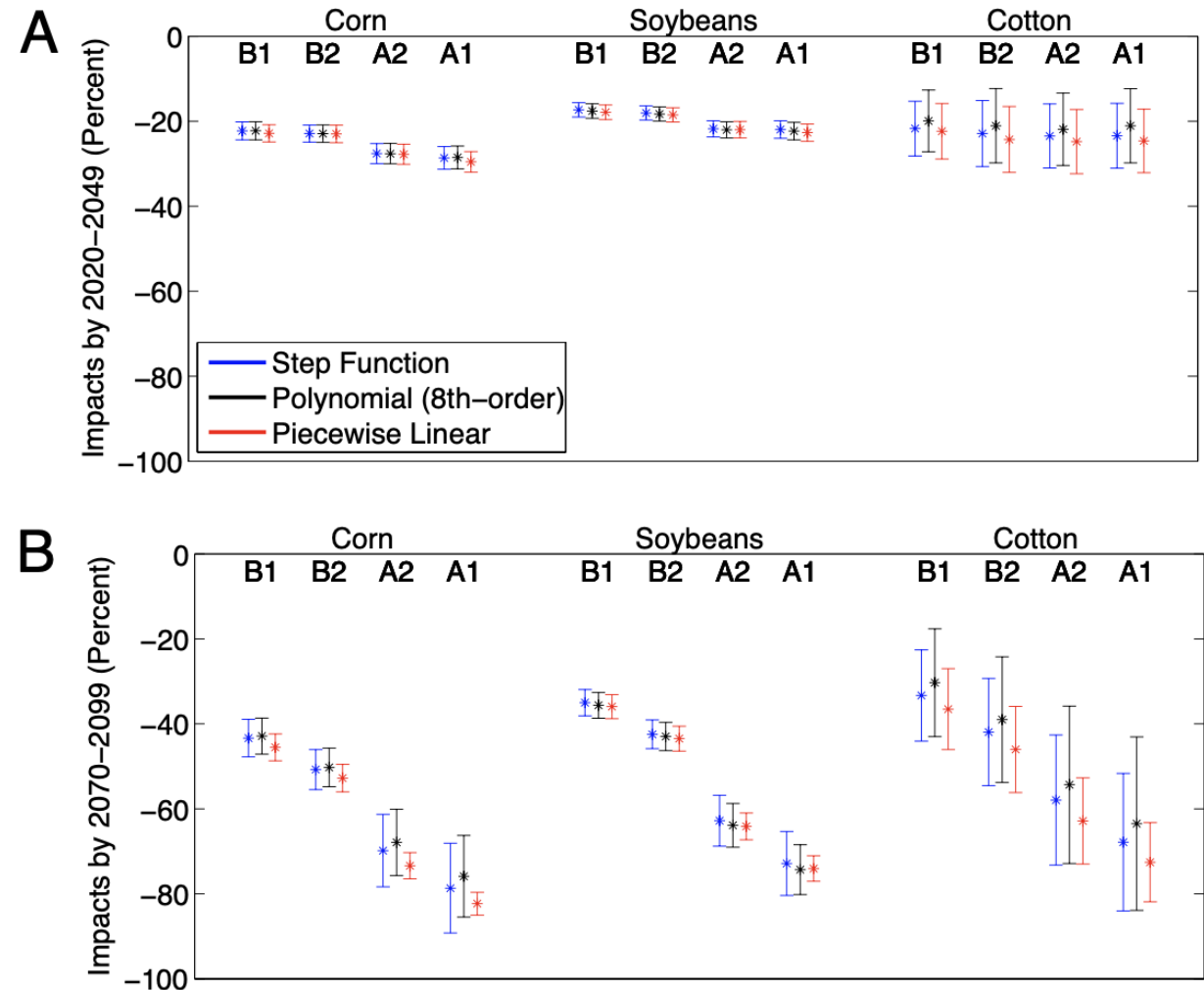
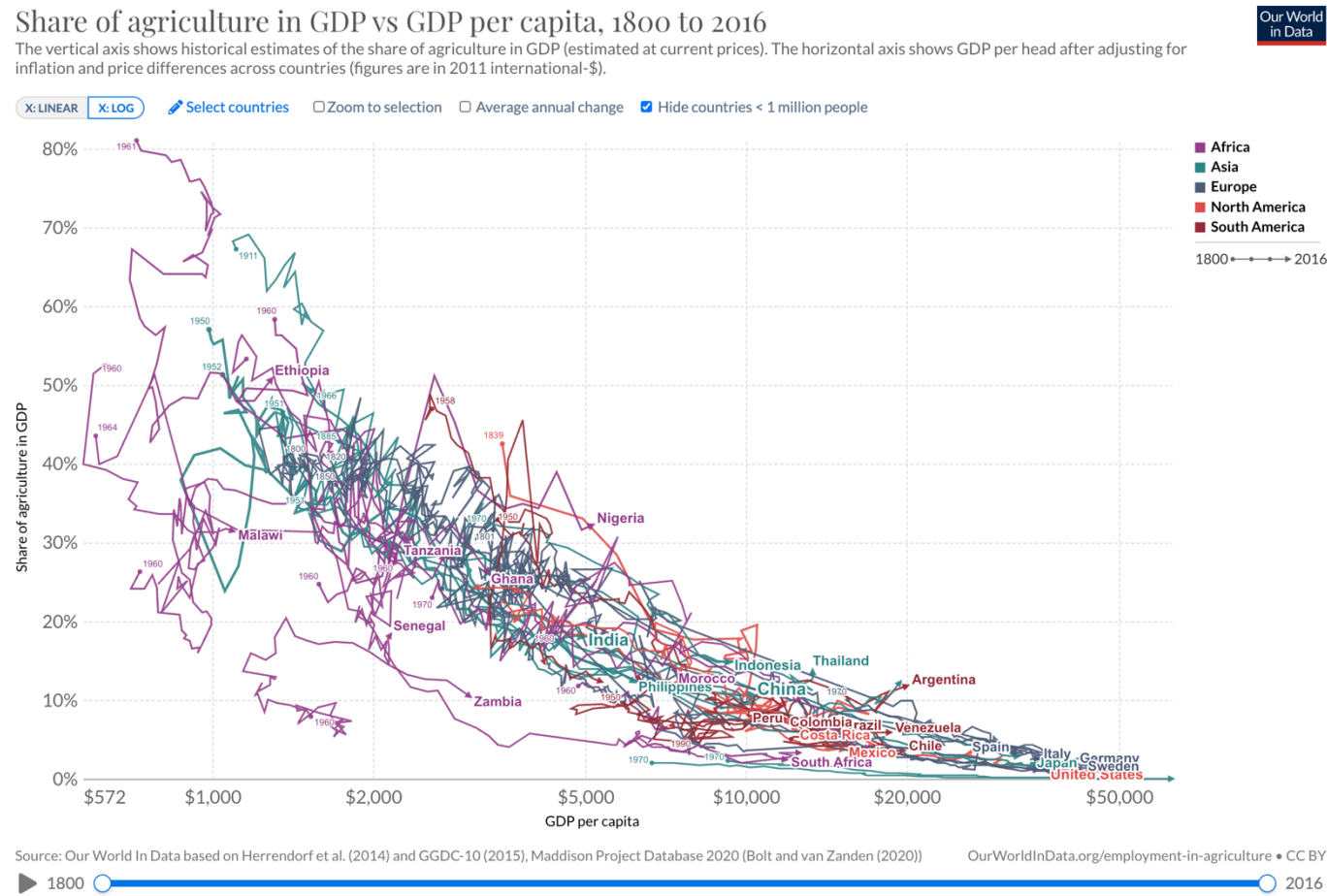


Fig. 2. Predicted climate-change impacts on crop yields under the Hadley III climate model. Graphs display predicted percentage changes in crop yields under four emissions scenarios. Frame A displays predicted impacts in the medium term (2020–2049) and frame B shows the long term (2070–2099). A star indicates the point estimates, and whiskers show the 95% confidence interval after adjusting for spatial correlation. The color corresponds to the regression models in Fig. 1.

Takeaways

- Schlenker & Roberts (2009) is one of the first in the “modern era” of the empirical climate impacts literature.
- Because agriculture is one of the central “market goods” affected by climate change, this structure forms the foundation for some of the “nonlinear” impacts research that follows (including the Burke, Hsiang, Miguel (2015) paper).
- Main takeaway is that temperature *extremes* are bad, particularly for agriculture, but for other things measured by GDP as well.
- Agriculture is a small component of US GDP (~2%) [larger component for developing countries] but global impacts of agricultural yields can be large.



As countries become wealthier, agriculture comprises a smaller and smaller portion of economic output. Source: [\[link\]](#)

For next week...

[http://www.caseyjwichman.com/econ4210 - week7](http://www.caseyjwichman.com/econ4210-week7)